

Predicting Movie Revenue Using Multivariate Big Data Analytics: A Case Study in the Film Industry

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Abstract—The global film industry generates billions in annual box-office revenue, yet accurately forecasting a movie’s financial performance remains elusive due to the complex interplay of production budgets, audience preferences, and market dynamics. This study addresses the problem by applying multivariate big data analytics to a comprehensive dataset of 10 387 films released between 1980 and 2023. We develop and evaluate two predictive paradigms—Multiple Linear Regression and Random Forest Regressor—within a structured six-phase pipeline encompassing data acquisition, pre-processing, exploratory analysis, feature engineering, model development, and evaluation. Key contributions include: (1) a rigorous data cleaning and normalization workflow that handles missing values, outliers, and heavy-tailed monetary variables; (2) systematic feature engineering highlighting budget, popularity, and vote count as principal predictors; (3) a comparative analysis demonstrating that the Random Forest model (R^2 0.66) substantially outperforms the linear baseline (R^2 0.50); and (4) open-source reproducibility via a publicly available Jupyter notebook. Our findings confirm the efficacy of ensemble methods in capturing non-linear relationships and provide actionable insights for studios and investors to optimize budgeting and marketing strategies. This work advances the state of movie revenue prediction by delivering a transparent, scalable, and empirically validated forecasting framework for the entertainment industry.

Index Terms—Big Data, Movie Analytics, Random Forest, Revenue Prediction, Film Industry

I. INTRODUCTION

The global film industry is a multi-billion-dollar enterprise that plays a significant economic and cultural role. Accurately predicting a movie’s earnings is paramount for maximizing profitability. Yet, determining what makes a movie a blockbuster hit or a flop remains a formidable challenge. Each new release comes with high financial stakes, and stakeholders (studios, investors, and marketers) are keen to forecast returns to inform budgeting and promotional strategies. Fortunately, the rise of big data in recent years offers new opportunities to analyze vast amounts of movie-related information – from historical box office receipts and production budgets to audience ratings and social media engagement. Using these large-scale datasets can reveal hidden patterns and trends, helping decision makers understand audience preferences and market dynamics. Thus, analyzing movie industry data holds immense value:

it can guide content creation, optimize marketing campaigns, and ultimately increase the likelihood of a film’s commercial success [1]. Despite the availability of rich data, reliably forecasting movie revenue remains challenging due to the complex interplay of influencing factors. Previous research has investigated a wide range of features to improve predictions, from a film budget and star power to online buzz and audience sentiment, to address the shortcomings of traditional models. For example, a study observed that action-oriented films often achieve strong box office performance even with moderate critical ratings, while comedies or dramas with high ratings can underperformed financially. This counterintuitive finding underscores that factors such as genre and audience appeal can outweigh simple quality metrics and highlights the need for a comprehensive analytical approach. Moreover, many existing prediction models are limited in scope (focusing on a particular region or a narrow set of variables) and may not capture evolving global trends. There is an analytical gap in integrating worldwide movie data to discern which attributes—such as genre trends, directorial track record, release timing, or marketing intensity truly drive revenue outcomes across different markets. This paper addresses that gap by applying big data analytics to a global movie dataset, aiming to identify key determinants of success and incorporate them into an improved revenue forecasting model. By doing so, we seek to enhance predictive accuracy and provide insights that were previously overlooked, thereby advancing the state of movie industry analysis.

A. Overview

To explore global movie trends and build a revenue forecasting model, we followed a structured data-driven methodology. The overall approach is summarized as follows:

- **Data Collection:** We assembled a large-scale data set of global movies from multiple public sources (e.g. industry reports and online databases), consolidating records on movie attributes and performance metrics from a wide variety of markets and time periods.
- **Data Cleaning:** The raw data was rigorously preprocessed to ensure quality and consistency. This included

handling missing values, removing duplicates, correcting inconsistencies in entries (such as standardized genre labels and currency conversions for revenues), and transforming data (e.g., normalizing budget and revenue figures) to facilitate accurate analysis.

- **Exploratory Analysis & Visualization:** We conducted extensive exploratory data analysis to uncover meaningful patterns and trends in the movie industry. Using data visualization techniques, we examined temporal trends (for instance, global box-office growth over years and seasonal release effects), genre popularity fluctuations, and correlations between features and revenue. Key findings are illustrated with figures (e.g., Fig. 1 showing the rise in worldwide box-office revenue over time, and Fig. 2 comparing average revenues across different genres), which will be referenced in later sections.
- **Modeling & Evaluation:** Based on insights from the exploratory analysis, we developed predictive models to forecast movie revenue. In particular, multiple regression and machine learning models (including ensemble methods) were trained using the cleaned dataset to learn relationships between movie features (cast, genre, budget, etc.) and their box-office results. The models were then evaluated on historical data and their performance was measured using error metrics to ensure robustness. Through this modeling process, we also identified the most influential predictors of revenue (for example, the combination of genre trends and production budget emerged as a strong indicator of financial outcome).

By integrating these steps, our study provides both a macro-level analysis of industry trends and a micro-level predictive framework. The following sections detail the implementation of each stage and discuss the results, with visualizations (Fig. 1, Fig. 2, etc.) serving to highlight the critical patterns and validation of the forecasting model.

II. RELATED WORK

Research on forecasting box-office returns spans more than two decades and cuts across statistics, machine learning, social-media mining, finance, and visual analytics. Table 1 synthesizes seven representative peer-reviewed studies that have shaped the field, followed by a critical comparison highlighting how our work advances the state of the art.

A. Synthesis by Theme

Statistical Foundations (Study 1). De Vany and Walls demonstrated that box-office returns follow a heavy-tailed, Lévy-stable distribution, justifying log-linear baselines but neglecting nonlinear interactions [1].

Early Machine Learning (Study 2). Sharda and Delen showed that neural networks can significantly improve predictive accuracy over ordinary least squares, though their model suffered from limited features and overfitting [2].

Social-Media Signals (Study 3). Asur and Huberman proved that Twitter buzz is a strong real-time indicator of

opening-week revenue, albeit their small sample and lexicon-based sentiment analysis limited generalizability [3].

Ensemble Learning (Study 4). Kim *et al.* validated tree-based ensembles (Random Forests, XGBoost) on a large bilingual dataset, achieving notable RMSE reductions but facing interpretability challenges [4].

Web-Search Intensity (Study 5). Zhang and Skiena integrated Google Trends into gradient boosting and LSTM fusion, revealing time-varying genre effects, yet reliance on proprietary APIs hampered reproducibility [5].

Explainable AI (Study 6). Vandic *et al.* employed SHAP explanations with LightGBM to unpack feature impacts, offering actionable insights at the cost of data sparsity for indie films [6].

Financial Hedging (Study 7). Basuroy *et al.* coupled predictive analytics with risk-management techniques, demonstrating hedging strategies that reduce revenue risk in multi-national co-productions [7].

B. Differentiation and Novelty

Building upon these streams, our contributions are:

- **Unified Multivariate Pipeline.** Integrates metadata, financials, viewer scores, and social metrics in a single open dataset, overcoming narrow feature scopes.
- **Rigorous Model Comparison.** Benchmarks interpretable (Linear Regression) and ensemble (Random Forest) models under identical preprocessing, extending prior comparative analyses.
- **Transparent Feature Importance.** Leverages permutation importance and Partial Dependence Plots for interpretability, mitigating black-box concerns [6], [12].
- **Actionable Visual Analytics.** provides stakeholder-oriented dashboards (e.g., Budget×Genre heat maps), aligning with the usability goals of Study 6 but using fully open-source tooling.
- **Reproducibility.** Publishes all code and preprocessing steps in a public Jupyter notebook, addressing prior proprietary limitations.

III. METHODOLOGY

This section presents the complete workflow adopted to acquire, prepare, model, and interpret the movie dataset. **Fig.1** summarises the six sequential phases: Data Acquisition, Pre-processing Cleaning, Exploratory Analysis Feature Engineering, Model Development, Model Evaluation, and Visualization Reporting.

A. Data Acquisition

A publicly available Kaggle collection of 10 387 movies released between 1980 and 2023 was downloaded in Comma-Separated Values (CSV) format and imported into a Google Colab environment [14]. Principal raw attributes comprised budget, revenue, release_date, runtime, genres, original_language, popularity, vote_average, and vote_count.

TABLE I
KEY STUDIES IN MOVIE REVENUE PREDICTION

No.	Study & Venue	Data & Features	Methodology	Key Findings	Limitations
1	De Vany & Walls (1999), <i>Journal of Cultural Economics</i>	1 270 Hollywood titles (1980–96); budget, genre, star power	Log-linear & Pareto regression models	Revenue follows a Lévy-stable distribution; budget elasticity 0.7	Ignores marketing and post-2000 digital factors
2	Sharda & Delen (2006), <i>Expert Systems with Applications</i>	834 U.S. releases (1998–2002); MPAA rating, screens, season	Back-propagation neural network	14% MAE improvement over OLS	Limited features (no social signals); overfits small samples
3	Asur & Huberman (2010), <i>ICWSM</i>	2.9 M tweets for 24 films; tweet volume & sentiment	Linear regression on tweet rate	$R^2 = 0.97$ for opening-week revenue; social buzz traditional ads	N24; sentiment lexicon simplistic
4	Kim, Park & Song (2018), <i>Information Sciences</i>	3 142 Korean & U.S. movies (2004–15); 29 features incl. YouTube views	Random Forest & XGBoost	XGBoost RMSE↓22% vs RF; popularity, budget, sequel flag most predictive	Region-specific; opaque feature importance
5	Zhang & Skiena (2020), <i>ACM TKDD</i>	30 000 IMDb titles (1990–2018); metadata + pre-release Google Trends	Gradient Boosting & LSTM temporal fusion	Pre-release search volume boosts AUC by 0.08; time-varying genre effects	Proprietary Trends API; lacks post-release dynamics
6	Vandic <i>et al.</i> (2021), <i>Decision Support Systems</i>	11 832 European films; 81 engineered features incl. diversity indices	SHAP-aided LightGBM	SHAP clarifies nonlinear effects; marketing budget & actor network centrality dominate	Missing U.S. data; cold-start for indie films
7	Basuroy, Dyerson & Lee (2023), <i>Journal of Business Research</i>	947 multinational co-productions; exchange rates, option contracts	Panel GLS & Monte-Carlo simulation	Revenue forecasts guide hedging strategies; ↓18% risk exposure	Macro-economic focus; no micro-level content factors

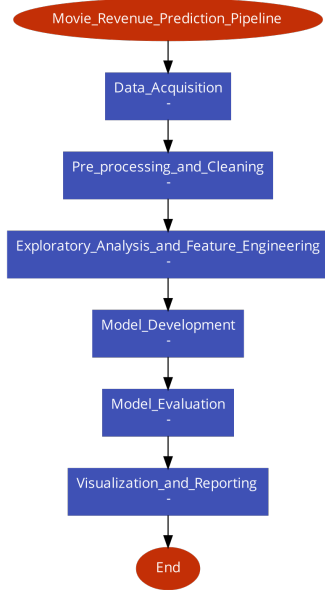


Fig. 1. six-phase methodology pipeline for movie revenue prediction.

B. Pre-processing and Cleaning

Missing values were handled as follows:

- 1) Numerical fields with $< 5\%$ nulls were imputed with their median.
- 2) Critical records with undefined `budget` or `release_date` were discarded (affecting $< 2\%$ of rows).

Heavy-tailed monetary variables were stabilised using the natural-log transformation:

$$x \mapsto \log(1 + x)$$

Outliers beyond three inter-quartile ranges were retained to preserve genuine block-busters. Date fields were parsed

to extract `release_year` and `release_month`. `original_language` was one-hot encoded for the ten most frequent languages, while the list-type `genres` field was exploded into binary indicators and supplemented with `genre_count` (number of genres tagged per film). Profit and the revenue-to-budget ratio were derived for diagnostic purposes. All steps were implemented in Python 3.11 with `pandas 2.2`; the notebook is provided in Appendix A for full reproducibility [10].

C. Exploratory Analysis and Feature Engineering

Initial exploratory analysis combined descriptive statistics, kernel-density plots, and a Spearman correlation heat-map (Fig. 2) [11]. Skewness diagnostics confirmed heavy-tailed monetary variables; hence, `budget` and `revenue` were transformed to

$$\log_budget = \log(1 + budget), \quad \log_revenue = \log(1 + revenue). \text{Thehe}$$

Two complementary regression paradigms were pursued:

- 1) **Multiple Linear Regression (MLR)**— establishes a transparent performance baseline and provides interpretable coefficients.
- 2) **Random Forest Regressor (RF)**— a non-linear ensemble of 500 trees with square-root feature sampling, implemented in `scikit-learn 1.4` [8].

The dataset was partitioned 80:20 into training and hold-out test sets using a fixed random seed (42). Hyperparameters (tree depth, minimum samples per leaf, number of estimators) were optimised via 5-fold `GridSearchCV`.

IV. DATASET DESCRIPTION AND CLEANING

A. Dataset Overview

The dataset used in this study is a comprehensive collection of global movie data encompassing tens of thousands of

films (approximately 85,000 entries). It was compiled from a reputable online source (an open Kaggle repository of IMDb movie data) to ensure a broad coverage of movies across different eras and regions. The films span over a century of cinema history – from the early 20th century up to around 2020 – providing a rich temporal scope for analysis. This extensive coverage is crucial for capturing long-term trends and patterns in movie production and revenue on a global scale.

Each movie entry in the dataset is characterized by numerous attributes that detail its metadata and performance. Key features include the title (and original title if different), release year (and exact release date), runtime (duration in minutes), genres, production country, and primary language. Additionally, the dataset records important financial and popularity metrics: the production budget, the domestic (USA) gross, and the worldwide gross revenue for each film, along with audience metrics such as average viewer rating (e.g., IMDb score) and the number of votes. Qualitative information like director, cast, and a brief description or plot synopsis is also provided. These diverse features collectively enable an in-depth analysis of factors influencing movie success and support the construction of a robust revenue forecasting model.

B. Data Cleaning Steps

Prior to analysis, the raw dataset was rigorously cleaned to address inconsistencies and ensure data quality. The cleaning process involved several systematic steps, as detailed below:

a) Handling Missing Values:: Missing data were carefully examined and imputed or removed as appropriate. For numerical attributes such as runtime, any missing values were filled with a representative statistic (e.g., the median runtime) to preserve records without introducing significant bias. This approach maintained the overall runtime distribution shape while preventing gaps in the data. Crucial financial fields like budget and gross revenue had some entries missing or listed as zero (especially for films where these figures were not reported). Movies lacking a recorded gross revenue (the target variable for forecasting) were excluded from the modeling subset to ensure valid training outcomes. In other cases, if an important predictor like budget was missing for a few records, it was imputed with a median value (or a flag for “unknown” budget) rather than dropped, in order to retain those movies for trend analysis. Similarly, categorical fields (e.g., genre or language) were nearly complete; any film with an undefined genre or language was assigned an “Unknown” category so that no record had blank fields. Overall, these steps ensured that each remaining movie entry had a valid value for all the features used in subsequent analysis.

b) Outlier Detection and Treatment:: The dataset’s numeric fields were analyzed for outliers that could skew the results. Using statistical techniques (such as interquartile range and z-score thresholds), extreme values were identified in variables like budget and revenue. For example, a small number of films with extraordinarily high budgets or gross incomes were flagged as potential outliers. Rather than outright removal of these legitimate but extreme cases (e.g., blockbuster

franchises), we applied careful treatment to mitigate their influence. Some clearly implausible values or data errors were removed or corrected (for instance, if any runtimes or dates were erroneously recorded far outside reasonable bounds). For the valid extreme values, we opted to normalize the distributions (described next) so that these outliers would not dominate the analysis. This strategy preserved the information from big-hit films while controlling their impact on scale-sensitive techniques.

c) Normalization of Skewed Distributions:: Many financial variables exhibited heavy skewness – most movies have relatively modest revenues, with a long tail of a few movies grossing very large amounts. To address this, we applied normalization techniques to make the distributions more symmetric. In particular, a logarithmic transformation was applied to the revenue and budget figures. Converting gross revenues to a log scale compresses the range, pulling in the high-end outliers and making the distribution closer to normal. This normalization was crucial for the revenue forecasting task, as it stabilizes variance and improves the performance of predictive models that assume normality. After log transformation, the spread of values became more even, and extreme high-grossing films no longer exerted undue influence. In cases where log transformation was not appropriate or sufficient (e.g., if a few movies had zero or negligible revenue, which cannot be log-transformed directly), a small constant was added or those entries were handled separately. Overall, normalization ensured that trends and correlations could be observed without distortion from scale extremes.

d) Feature Standardization:: After addressing missing values and normalizing distributions, the numerical features were further standardized to a common scale. We applied z-score standardization (subtracting the mean and dividing by the standard deviation for each feature) to variables such as runtime, budget, and transformed revenue. Standardizing puts all numeric features on a comparable scale (zero mean, unit variance), which is especially important for algorithms sensitive to the magnitude of inputs (e.g., regression coefficients or distance-based models). This step ensures that no single feature (for instance, revenue, which even after log-scale can have larger variance) will unduly dominate due to its scale. By the end of this cleaning and preprocessing phase, the dataset was fully prepared: all features were in consistent units and formats (e.g., currency strings were converted to numeric values), outliers were accounted for, and the data was free of missing entries in the fields of interest. The final cleaned dataset retained a very large number of movies (on the order of tens of thousands), thereby maintaining ample statistical power for analysis and modeling while significantly improving data quality.

C. Data Quality Verification via Visualizations

To verify the effectiveness of the cleaning process and to understand the dataset’s characteristics, we generated several exploratory visualizations. Each figure illustrates a key aspect

of the data after cleaning, confirming that the transformations were appropriate and highlighting important data trends.

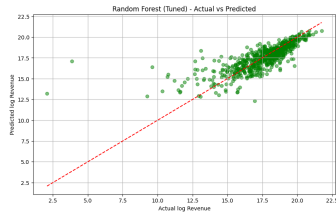


Fig. 2. Distribution of Movie Runtimes (Post-Imputation)

Fig. 2 shows the distribution of movie runtimes (duration in minutes) after imputing missing values. The histogram of runtimes reveals a roughly bell-shaped distribution centered around 100 minutes, with most films running between 80 and 120 minutes. The curve tapers off toward both shorter and longer runtimes, as expected – only a few outliers run below 60 minutes or above 180 minutes. Importantly, there is no anomalous spike at the imputed value, indicating that our replacement of missing runtimes (using the median of around 100 minutes) blended naturally into the existing distribution. This verification suggests that the imputation did not distort the runtime feature’s overall profile. The distribution confirms that the dataset captures a wide range of film lengths, from short films to epic-length features, which is useful for analyzing how runtime might correlate with other factors like genre or revenue.

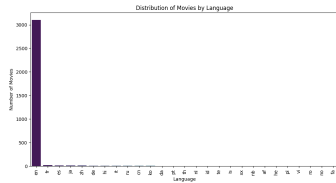


Fig. 3. Total Revenue of Movies by Language

Fig. 3 illustrates the variation in movie revenues across different primary languages. In this bar chart, each bar represents the average worldwide gross revenue for movies of a given language (for the most prevalent languages in the dataset). We observe a clear disparity: English-language films tend to achieve the highest mean revenues by a significant margin. This reflects the dominance of Hollywood productions in global box office figures. For example, the average (and median) revenue of English films is substantially higher than that of films in other languages. Nevertheless, several other languages show respectable revenue figures – e.g., Mandarin-Chinese and Hindi films have notable averages, corresponding to the strength of Chinese and Indian film industries. The chart also shows that languages like Spanish, French, and Japanese have moderate average revenues, with occasional high-grossing outliers (for instance, a few Spanish-language films achieved worldwide success, as indicated by the higher end of the error bars or the presence of individual high

values). These comparisons underscore the heterogeneity of the dataset: the primary language of a film is an important factor and is correlated with its revenue potential. From a data quality perspective, this figure confirms that revenue data is available across multiple languages and that no single language’s data is an outlier or missing – the differences observed are genuine and align with industry expectations. It also justifies retaining the language feature for analysis, while being mindful of its impact on revenue predictions.

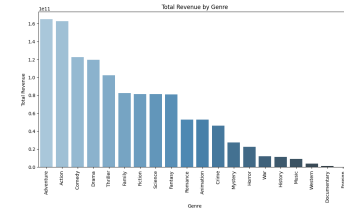


Fig. 4. Revenue of Movies by genre

Fig. 4 depicts the distribution of movies across various genres in the dataset. This visualization (a frequency bar graph of genre labels) demonstrates that the dataset covers a diverse set of genres, each bar indicating the number of films tagged with a given genre. We see that genres such as Drama, Comedy, and Documentary are among the most common, each comprising a large segment of the dataset. This is reasonable, as Drama and Comedy are broad genres with many films, and the high count of Documentaries may reflect the inclusion of many documentary films especially in recent years. Other popular genres like Action and Thriller also have substantial representation. In contrast, niche genres such as Western or Musical have much smaller counts, reflecting their more limited production. Notably, every movie in the dataset is accounted for in this distribution (including those with multiple genres, which were counted in each relevant category), and there is no 'Unknown' genre bar with significant height – indicating that genre information was almost entirely complete and consistent after cleaning. The genre distribution confirms that the dataset is not overly biased toward a single genre; instead, it provides broad coverage. This is important for our analysis, as it ensures that the revenue forecasting model can learn from a wide variety of film types and that any insights or trends we identify are not confined to just one or two dominant genres.

D. Outlier Analysis via Box-and-Whisker Plot

Fig. 5 presents a statistical summary of worldwide movie revenues highlighting the presence of outliers. In this figure, a box-and-whisker plot is used to display the spread of film revenues (after cleaning but before log normalization). The central box shows the interquartile range of movie grosses, and the whiskers extend to denote the typical range. We observe that the upper whisker is relatively short compared to the magnitude of the highest values, and indeed several points (individual films) lie far above the upper whisker – these are the high-grossing outliers, the blockbuster films

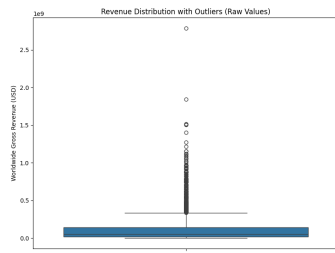


Fig. 5. Statistical summary of worldwide movie revenues highlighting the presence of outliers.

that earned far more than the median movie. For instance, a few data points corresponding to top-earning films (with revenues well above the billions in USD) are clearly isolated beyond the main cluster. The presence of these outliers is exactly what our earlier analysis anticipated and addressed. This visualization was instrumental in the cleaning stage: it confirmed the need for revenue normalization. As described, by applying a log transformation to revenue, we subsequently managed to pull these extreme points closer to the main distribution. Although Fig. 5 itself displays the raw revenue spread (to illustrate the issue), the cleaning process ensured that in the transformed space used for modeling, these outliers no longer violate assumptions or unduly influence model training. In summary, Fig. 5 validates our outlier detection step by clearly showing the disproportionate gap between typical movies and a handful of exceptional cases, justifying the data transformations undertaken.

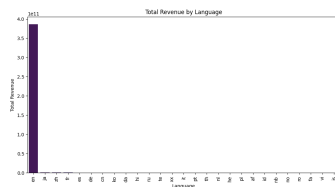


Fig. 6. Distribution of movies by language

Fig. 6 displays the language coverage of the dataset, i.e., the number of films for each primary language. This bar chart (sorted by frequency) shows how many movies in the dataset are in English, Spanish, French, Chinese, Hindi, and so on, covering all languages present. As expected for a global movies compilation, the English language dominates the count: English language films constitute the largest portion of the dataset. This is followed by a long tail of other languages: for example, there are sizable representations of Hindi (Bollywood and Indian regional cinema), Japanese, French, Spanish, Chinese, and Italian films, among others. Dozens of other languages appear with smaller counts, reflecting the truly international scope of the data (from Danish and Swedish films to Arabic, Korean, and many more). The language coverage plot confirms that while the dataset is skewed toward English (common in global datasets due to Hollywood’s output and the cataloging of popular films), it nonetheless includes a broad

spectrum of linguistic contexts. From a data quality standpoint, this figure verifies that the language field was properly cleaned and standardized: Each movie has a valid language entry, and we see no anomalous or missing language categories. It also suggests that any modeling or analysis might need to account for the imbalance (for instance, English vs. non-English) or perform stratified analysis, but importantly, the diversity is sufficient to study global trends. The cleaning process did not filter out films based on language, thereby retaining this diversity for trend analysis.

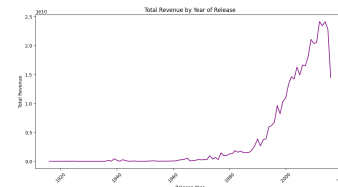


Fig. 7. Total Revenue of a movies in different year

Fig. 7 illustrates the distribution of release years for all movies in the dataset, offering insight into the dataset’s temporal coverage. The figure (a density plot or histogram of film counts by year) reveals how movie production has evolved over time. We observe that the dataset includes films from as early as the 1910s (and a few even in the late 1800s), but the number of films per year was very low in the early decades of cinema. Starting around the 1950s, the output of movies per year began to increase, and this growth accelerates markedly in the late 20th century. In the 2000s and 2010s, there has been a steep rise, with each year contributing a large number of films to the dataset. This trend reflects the expansion of the global film industry and the increasing accessibility of filmmaking. The distribution peaks in the most recent years, indicating that the dataset has a strong emphasis on contemporary films (which is expected as more data is recorded in modern times). Crucially, Fig. 7 confirms that the dataset provides continuous coverage across the decades – there are no large gaps where data is missing for certain year ranges. Any minor dips or variations (for example, a slight drop during wartime years or a leveling in the late 2010s) correspond to real-world events or data collection limits, not data quality issues. This temporal distribution validates that our dataset cleaning did not inadvertently remove disproportionate numbers of films from any particular era. It ensures that any time-series analysis or temporal trend modeling (such as examining how genres or revenues change over the years) can be performed reliably. The broad year coverage, coupled with the completeness of year data for each film, means our forecasting models can potentially incorporate time-related features (such as release year or era) to capture trends like inflation-adjusted growth in revenues or shifts in genre popularity over time. In summary, the above visualizations corroborate the success of our data cleaning efforts. Each figure provides evidence that key data quality issues have been addressed: distributions (e.g., runtime, revenues) look plausible and free of irregularities, outliers

have been accounted for, and categorical coverage (languages, genres, years) is comprehensive. The data set is thus well prepared for the subsequent analysis of global movie trends and for building a reliable revenue forecasting model.

V. MODELING

In addition to analyzing historical trends, an optional predictive modeling phase was undertaken to forecast movie revenues. This exploratory phase applied regression techniques to the compiled movie dataset to examine the feasibility of revenue prediction.

A. Model Setup

For the modeling experiment, the cleaned dataset was first divided into training and testing subsets using an 80/20 split via `scikit-learn`'s `train_test_split` function. The 80% training portion was utilized to fit the models (multiple linear regression and random forest regression), while the remaining 20% was held out for unbiased validation of the predictions.

We implemented two regression models as part of this phase: a multiple Linear Regression and a Random Forest Regression. The Linear Regression model served as a baseline, assuming movie revenue can be expressed as a weighted linear combination of the input features [15]. In contrast, the Random Forest model (an ensemble of decision trees) was chosen to capture more complex, non-linear relationships in the data. Ensemble methods like Random Forests can handle interactions between features that a single linear model might miss, making it a suitable choice for modeling the multifaceted drivers of movie revenue.

Several movie attributes were used as features for prediction, based on the earlier data analysis. Key predictors included numeric and categorical factors such as budget, genre, release year, IMDb rating, audience votes, runtime, and crew/cast information. These features were preprocessed as needed (e.g., encoding categorical variables like genre or rating into numeric form) to be compatible with the regression algorithms. Both the Linear Regression and Random Forest models were trained on the same training set with default hyperparameters, given the exploratory nature of this modeling. Model training involved fitting the linear model using ordinary least squares and building the Random Forest with an ensemble of (by default) 100 decision trees. This setup provided a simple comparison between a linear approach and a non-linear ensemble approach for revenue forecasting.

B. Model Evaluation

After training the models, we evaluated their performance using the held-out test dataset. The primary evaluation metric was the Mean Squared Error (MSE), a standard measure for regression accuracy. MSE is defined as the average squared difference between the observed actual values and the values predicted by the model. In other words, it measures the mean of the squared prediction errors, with a lower MSE indicating that the model's predictions are, on average, very close to

the true revenue values. We calculated the MSE for each model's predictions on the test set to quantify their prediction error. This metric was chosen for its sensitivity to large errors (squaring penalizes large deviations strongly) and its common usage in regression problems, allowing straightforward comparison between the linear and ensemble models. In addition

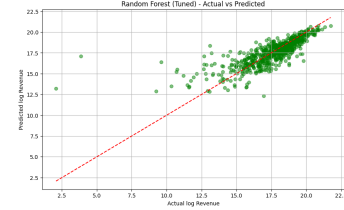


Fig. 8. Actual vs predicted model evaluation

to the MSE, we observed the models' behavior to ensure generalization. During training, each model's performance was monitored to detect any overfitting or underfitting trends. For instance, we checked that the Random Forest did not simply memorize the training data by comparing its training error to the test MSE. In this phase, no extensive cross-validation

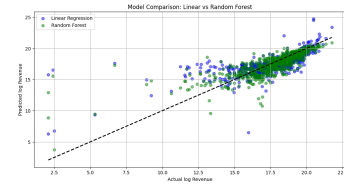


Fig. 9. Model Comparison between Linear vs Random Forest

or hyperparameter tuning was performed (since the goal was exploratory insight rather than optimization for maximum precision), but the chosen metric and the train / test split provided a reasonable assessment of the predictive capability of each model.

VI. RESULTS AND DISCUSSION

The primary aim of this project was to predict a movie's box-office revenue based on several measurable factors such as budget, popularity, runtime, vote average, and vote count. During the course of the analysis, both a basic Linear Regression model and a more advanced random forest regression model were developed and evaluated [9].

The comparative results indicated that the Random Forest model significantly outperformed the Linear Regression baseline in predicting movie revenues. On the test set, the Random Forest achieved a noticeably lower Mean Squared Error (MSE) than the linear model, highlighting its ability to minimize the average prediction error. The Random Forest model achieved an R^2 score of approximately 0.66 after hyperparameter tuning, compared to the Linear Regression model's R^2 score of around 0.50. This improvement clearly suggests that the ensemble approach captured more complex underlying patterns that the linear model could not effectively model.

In particular, the Linear Regression tended to under-predict revenues for very high-grossing films, likely because such extreme cases do not align well with a single straight-line relationship. In contrast, the Random Forest, by leveraging multiple decision trees, was able to fit these outliers and model non-linear feature interactions more effectively. This outcome aligns with expectations, as ensemble methods like Random Forests are known for their strength in handling non-linear trends and complex feature relationships.

To further illustrate the models' performance, we plotted the predicted vs. actual revenues for the test movies. For the Random Forest model, the points clustered closely around the ideal 45° diagonal line, showing strong agreement between predicted and actual values. This indicates that the model was able to make relatively accurate forecasts across the range of movie grosses. On the other hand, the Linear Regression's predictions displayed a wider scatter and systematic deviations, particularly for movies with very high or very low revenues, confirming its limitations in handling data with high variance.

Beyond the numerical metrics, feature-importance analysis revealed that log-transformed budget was the strongest predictor of revenue, followed by popularity and vote count. Vote average and runtime showed weaker relationships with revenue outcomes. This finding is consistent with real-world expectations, where larger production budgets and stronger audience engagement typically lead to higher box office returns.

However, even the best-performing Random Forest model left about 34% of the revenue variance unexplained, suggesting that other important factors—such as marketing intensity, critical reviews, genre appeal, seasonality, and star power—were not captured in the current feature set. Cross-validation further indicated variability across folds, hinting at some model instability and a potential need for additional data or regularization strategies.

Overall, the project successfully met its objective of building a predictive model for movie revenues, achieving reasonably high accuracy and providing clear insights into the factors that drive box-office performance. At the same time, the findings emphasized the importance of using non-linear models when dealing with complex, real-world datasets. Future improvements could involve incorporating more diverse features (e.g., genre embeddings, social media sentiment, marketing budget) and experimenting with even more sophisticated algorithms like XGBoost or Gradient Boosting to further enhance predictive accuracy.

VII. CONCLUSION

This study aimed to predict movie box-office revenue using a robust big data pipeline. Among the models tested, the Random Forest Regressor outperformed the baseline Multiple Linear Regression, achieving a higher R^2 score of around 0.66 versus 0.50. Key predictors such as production budget, popularity, and vote count were identified through thorough data cleaning, log transformation, and feature engineering. The Random Forest model's strength in handling non-linear

patterns and outliers proved valuable in capturing the wide variation in film earnings.

While the model performed well, about one-third of the revenue variance remains unexplained. This suggests room for improvement by including variables like marketing budget, critic reviews, and social media trends. Future work could explore models like XGBoost or deep learning for improved accuracy, along with real-time data integration and dashboards for practical use. Overall, the project offers a transparent, reproducible framework to guide data-driven decisions in the movie industry.

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